Birdcage Diagram

User Guide for the Interactive Demo

1. Overview

The Birdcage Diagram visualizes the evolution of hierarchical grouping structures across multiple time slices. Each slice shows the hierarchy at one point in time, drawn as a stack of horizontal *blocks*. *Bands* connect blocks across adjacent slices and reveal how groups split, merge, persist, or shift through the hierarchy over time.

2. Quick Start

Once you have installed the dependencies (see the README in the GitHub repository), launch the application from the command line by passing one or more slice files via the `--slice` flag:

```
python birdcage_diagram.py --slice slice1.xlsx --slice slice2.xlsx --slice
slice3.xlsx
```

Open the URL printed in the console (`http://127.0.0.1:8050/`) in your browser. From there you can:

- Switch hierarchy level using the **Category** combo box.
- Click on any block, band, or collapse handle to interact.
- Use the **Upload files** button on the toolbar to load a different dataset without restarting the server — either a single multi-sheet Excel file (each sheet becomes one slice) or several single-sheet files (each file becomes one slice; an ordering dialog will appear so you can reorder them before clicking **Generate**).
- Use the **Download** button to export the current diagram as PNG, SVG, or PDF.

3. Data Format

Each slice is a tabular file (`.xlsx`, `.xls`, `.csv`, or `.ods`) with one column per hierarchy level plus a rightmost **element** column. Column names are arbitrary — the rightmost column is always treated as the element identifier.

Example (one slice):

Sector	Subsector	Industry Group	NAICS Industry	Element
Accommodation and Food Services	Accommodation	Traveler Accommodation	Hotels and Motels	Hotels and Motels
Accommodation and Food Services	Accommodation	Traveler Accommodation	Casino Hotels	Casino Hotels
Accommodation and Food Services	Food Services	Restaurants	Full-Service Restaurants	Full-Service Restaurants

Rules:

- Blank cells in hierarchy columns inherit the value above (fill-down behavior).

- Element values must be unique within a single slice.
- Across slices the column set must match. Column order may differ — columns are auto-aligned to the first slice.

4. Toolbar Reference

Button	Function
Upload files	Load slice data. Accepts .xlsx, .xls, .csv, .ods. Supports both multi-sheet single files and multiple single-sheet files.
File details	Show metadata of the currently loaded slices: file names, row counts, and column schemas.
Category	Combo box that selects which hierarchy level is used as the diagram's category axis. Must not be the rightmost column.
Ordering	Open the slice-ordering dialog. Drag or use arrows to reorder slices, then confirm to re-render.
Download	Export the current diagram. Choose format (PNG, PDF, SVG), resolution, and dimensions, then confirm.
Guide	Open this user guide.

5. Canvas Interactions

Action	Effect
Click on empty canvas	Refresh the canvas and clear any active highlight. Clicking outside any block or band — anywhere on the canvas background or on the empty area surrounding the diagram — re-fits the layout to the viewport and deselects the currently highlighted component.
Click on a block	Highlight the block and trace all bands flowing into and out of it. Click again or click on empty canvas to clear.
Click on a band	Highlight the band's source block, target block, and the elements that participate in this transition.
Click on a collapse handle	Collapse a sub-tree of the hierarchy into its parent, merging blocks and their bands. Collapse is purely visual; the underlying data is unchanged.
Click on an expand handle	Re-expand a previously collapsed sub-tree.
Double-click on the canvas	Reset zoom and pan to the default fit-to-view.
Mouse-wheel / pinch on canvas	Zoom in and out around the cursor position.
Drag on canvas	Pan the view.
Ctrl + Double-click on UI	Reset the global UI scale (top toolbar size, font size).